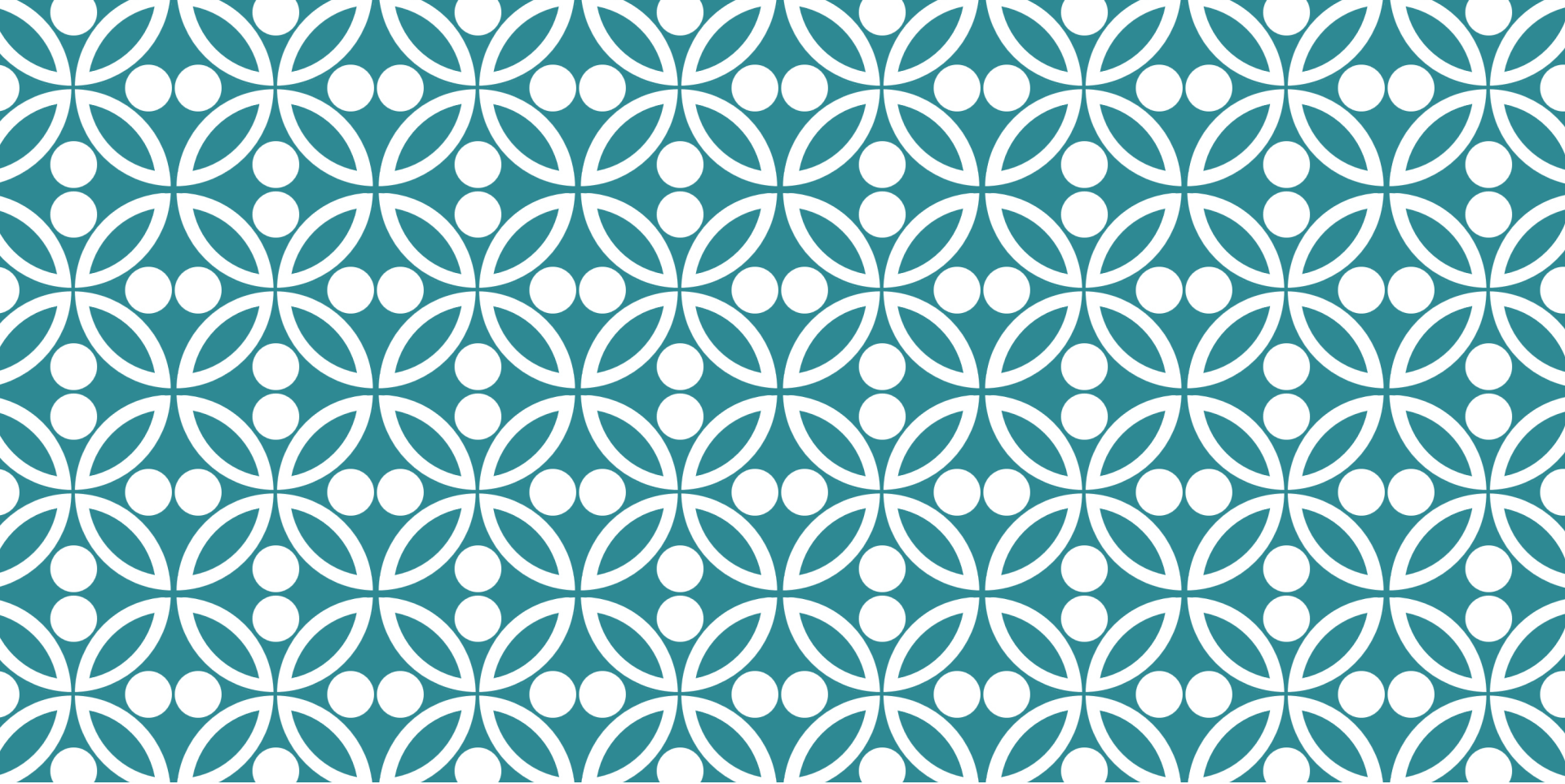




CLUSTER ANALYSIS

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CLUSTER ANALYSIS

Cluster Analysis is grouping a set of objects in such a way that objects in the same group (**cluster**) are more similar to each other than objects in other clusters.

Cluster Analysis:

- Generates similar groups
- Homogeneous within the group
- Heterogeneous as much as possible to other groups
- Data typically consist of objects or people

It is a general task, **not a specific algorithm**

Can be achieved by various algorithms- each differs significantly in the constitution of a cluster and how to find them

Formulated as a **multi-objective optimization** problem

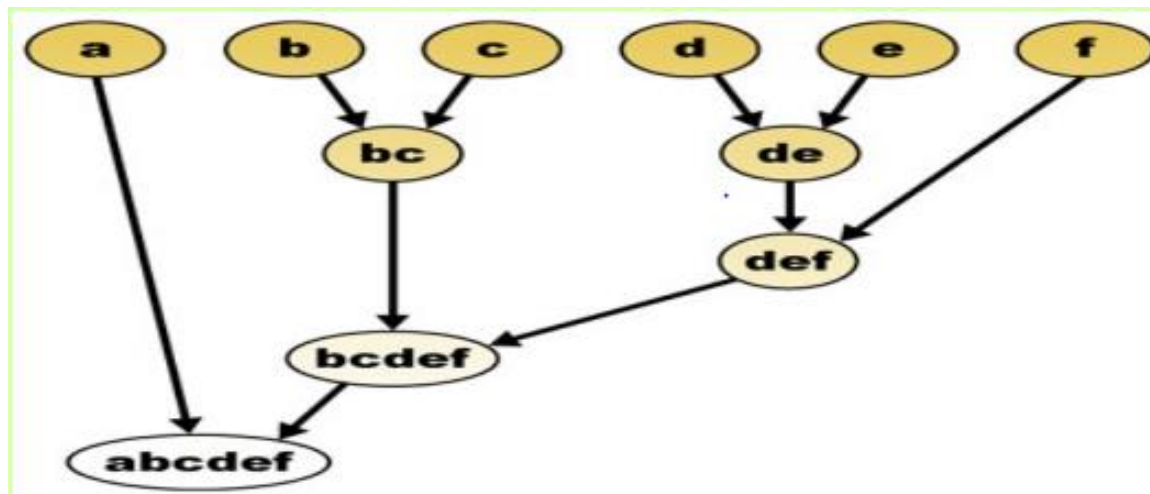
- Appropriate algorithm depends on the dataset and use of the results

EXAMPLES CLUSTER ANALYSIS DATASETS

Socio-economic criteria: income, education, age, profession

Psychographic criteria: lifestyle, values, interests, motivation

Ecological criteria: biome, lifestyle, physical characteristics



CONSTRAINED CLUSTER ANALYSIS

Constrained Cluster Analysis is a multivariate method for defining stratigraphic zones.

Stratigraphy- layers. Something must come in a specific order.

- Examples: geological layers, cake, building floors

It is **constrained** because only adjacent clusters are considered for merging.

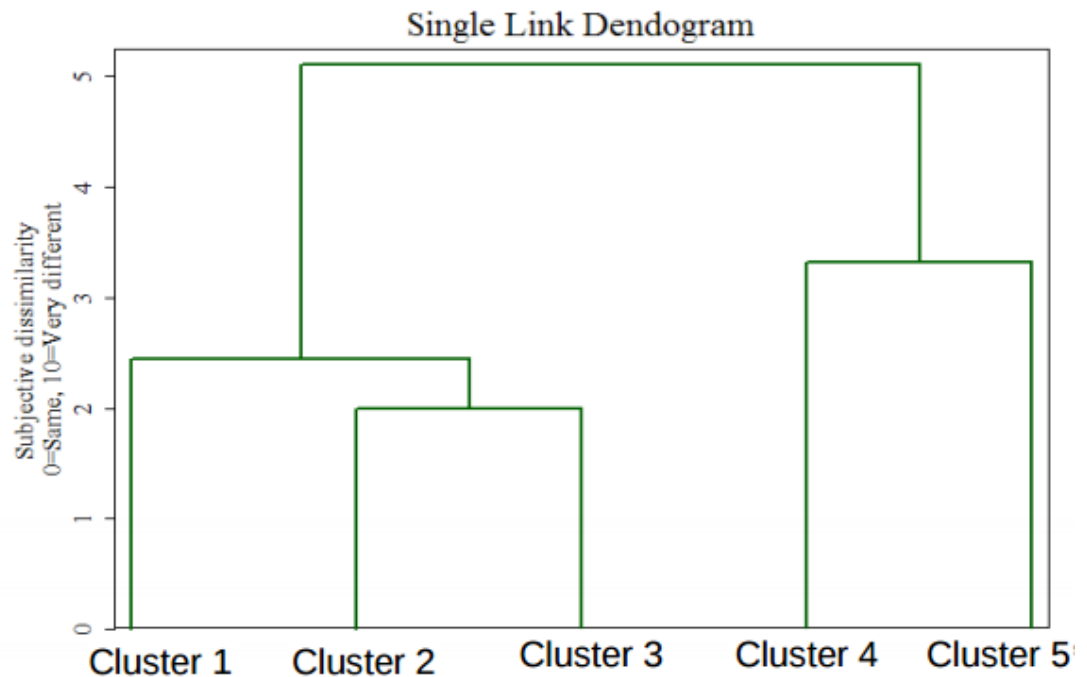


BASIC INTRODUCTION

Incremental sum of squares used with added data transformations

Two common methods:

- Splitting
- Agglomerative



SPLITTING

Attempt to divide the data sequence so:

- Sum of variances is reduced relative to total variance of complete dataset

Can be “**binary**” or “**optimal**”

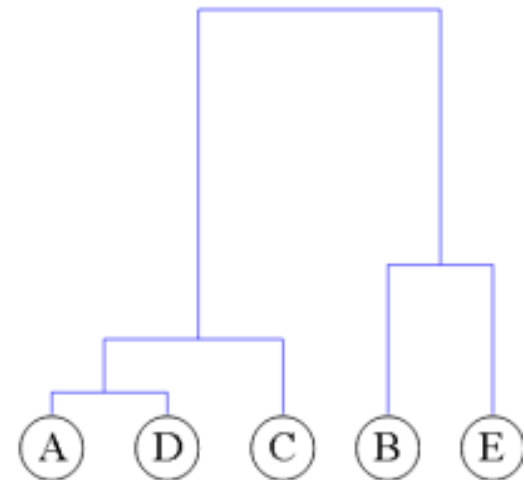
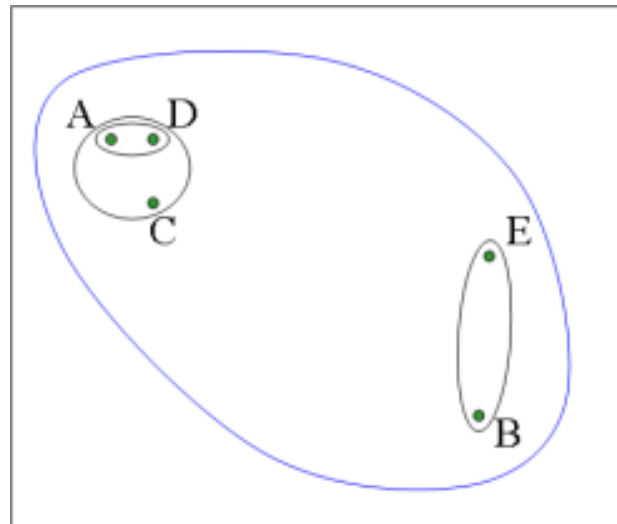
Binary: data divided into the best way possible, then making the best division of those zones, etc.

Optimal: data divided directly into the desired amount of zones

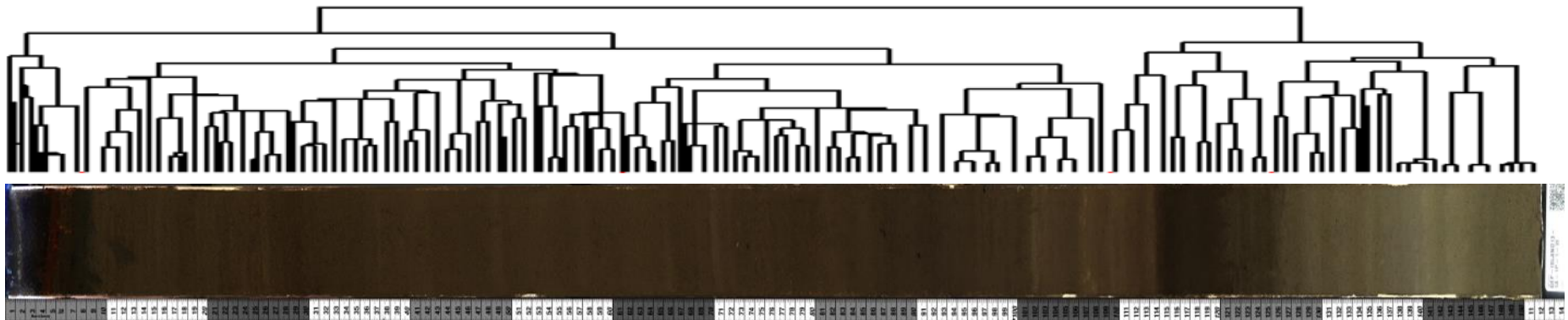
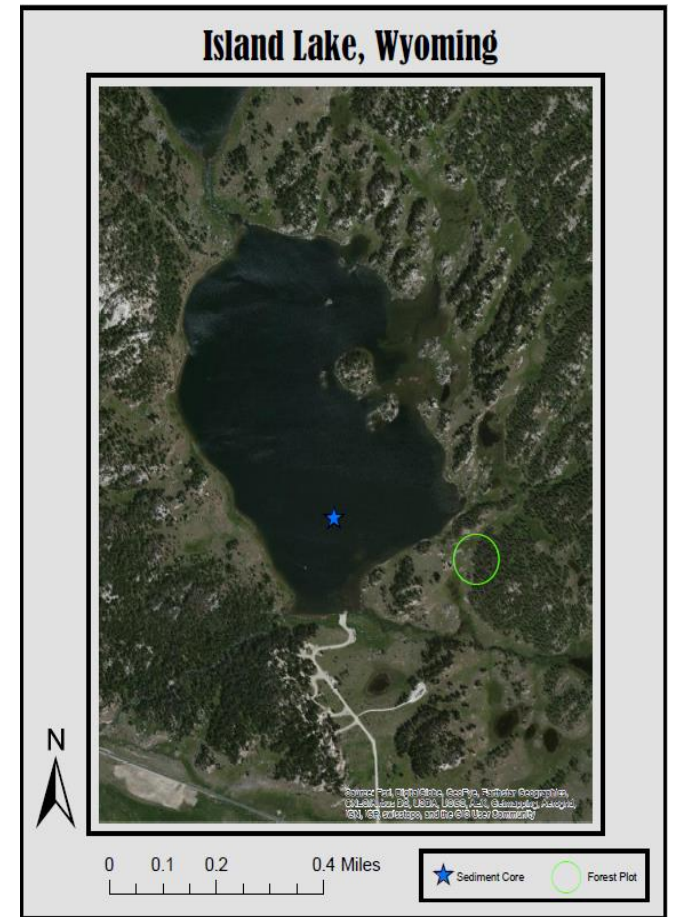
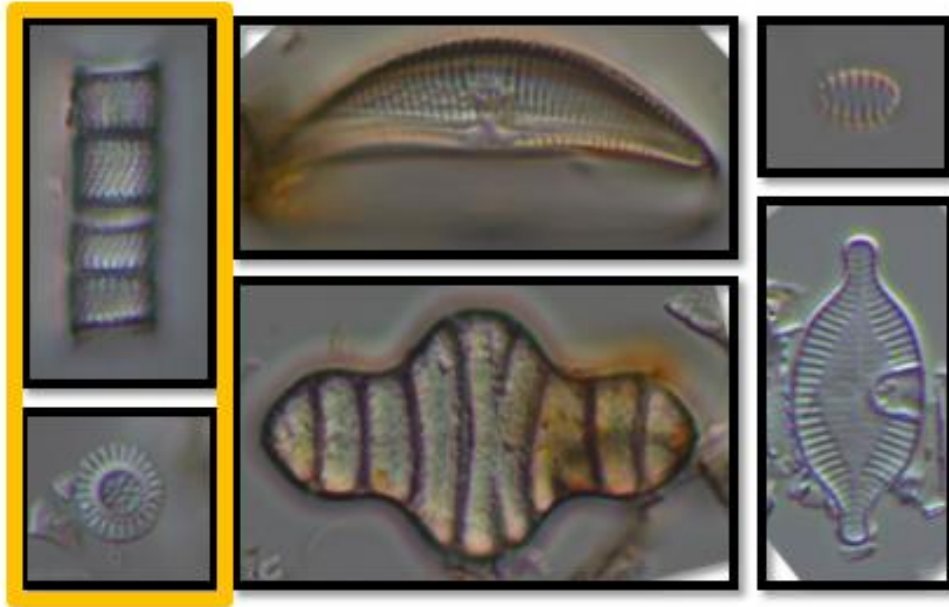
The best method is the one that minimizes the sum of variances at each stage.

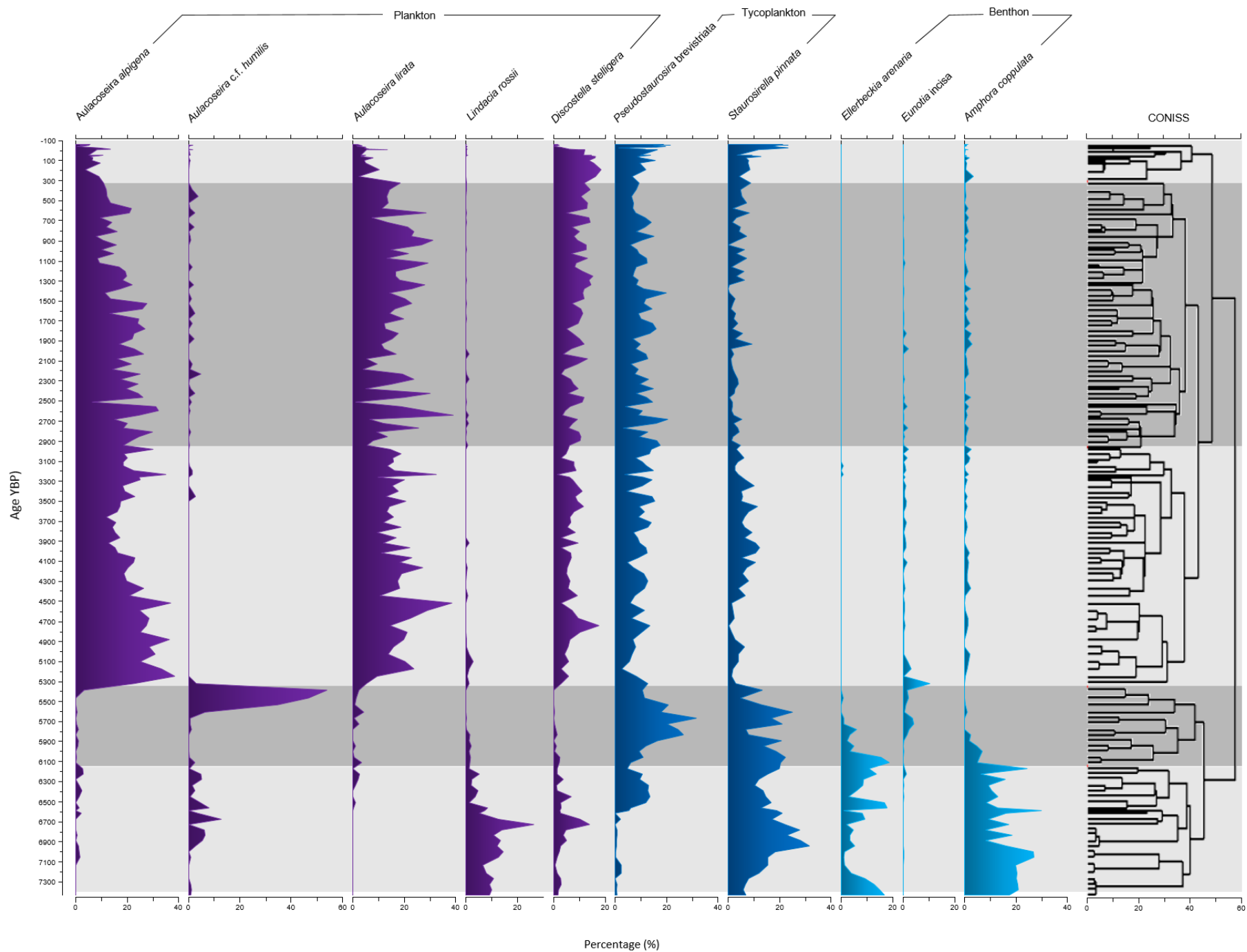
AGGLOMERATIVE

1. Searches the dataset for the two most similar, adjacent samples and combining
2. Combination treated as a single sample and process repeated



EXAMPLE: DIATOMS IN A SEDIMENT CORE

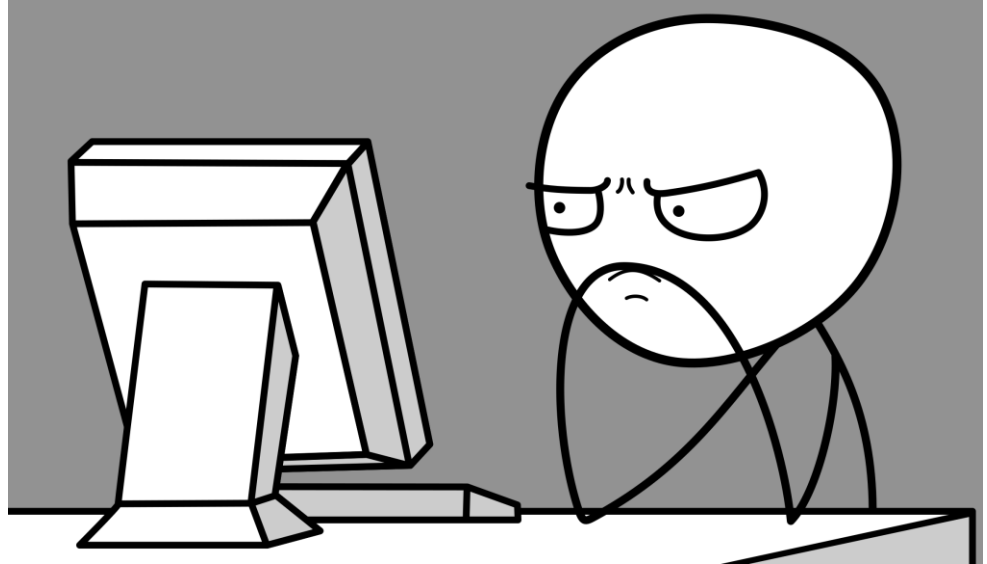




ZONATION SCHEME SOFTWARE

There are a number of software programs available to run cluster and constrained cluster analysis. We will briefly discuss:

- Psimpoll
- POLPAL
- ZONE
- Tilia



PSIMPOLL

Program for plotting and analysis of paleoecological data

Main features:

- High quality graphical outputs of stratigraphic data
- Wide range of numerical techniques
 - Zonation
 - Ordination
 - Basic statistical description
- Input and output is plain text
- Versions available on all computer operating systems
- PostScript graphical output can be modified further or converted

Available at: <http://www.chrono.qub.ac.uk/psimpoll/psimpoll.html>

POLPAL

Application for plotting and counting data as well as performing numerical analysis.

Data stored in tables with samples in rows

- MS Excel or NotePad

Free and Paid versions

Numerical Analyses

- CONSLink (Constrained Cluster Analysis)
- PCA (Principle Component Analysis)
- Minimal sum (excludes observations under a selected value)

Diagram exports as a bitmap file

Available at: http://adamwalanus.pl/polpal_manual.html#NumAnal

ZONE

MS-DOS program

Facilitates description and correlation between

Variety of statistical analyses:

- CONSLINK
- CONISS
- Binary splitting by least-squares
- Optimal splitting by least-squares
- Variable barriers approach

Available at: <http://www.chrono.qub.ac.uk/datah/zonation.html>

TILIA

Common program for plotting and analyzing stratigraphic data

Can be used on a variety of stratigraphic, geologic, and ecological data

High degree of control over diagram format and appearance

Runs on Windows

Free full student license

CONISS- Constrained Incremental Sums of Squares cluster analysis

Available at: <http://www.chrono.qub.ac.uk/datah/tilia.html>

SUMMARY

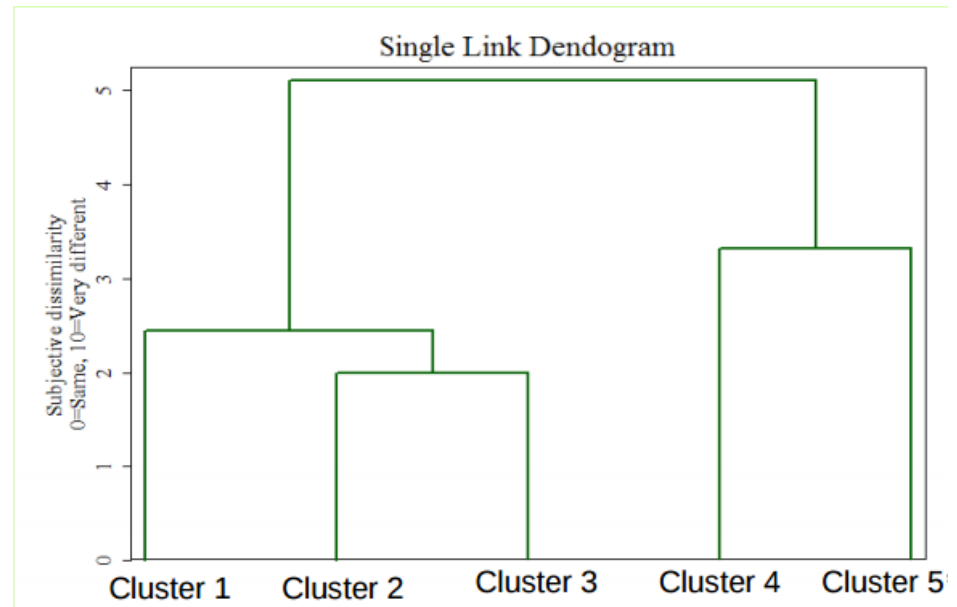
Cluster Analysis- method to determine similar groups

Constrained cluster analysis- stratigraphic clustering

- Splitting
- Agglomeration

Example

Zonation Scheme Software



SOURCES

- [http://www.focus-balkans.org/res/files/upload/file/9%20Cluster Analysis%20Schaer.pdf](http://www.focus-balkans.org/res/files/upload/file/9%20Cluster%20Analysis%20Schaer.pdf)
- <http://www.chrono.qub.ac.uk/datah/zonation.html>
- <http://www.chrono.qub.ac.uk/psimpoll/psimpoll.html>
- http://adamwalanus.pl/polpal_manual.html#NumAnal
- <http://www.chrono.qub.ac.uk/datah/tilia.html>